Cops and Robbers: A Graph Theory Game

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"Mathematics is the queen of the sciences and number theory is the queen of mathematics."
— Carl Friedrich Gauss

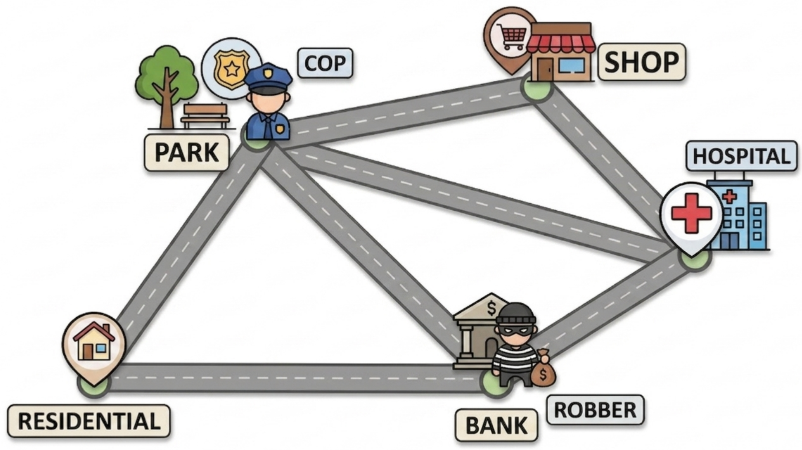
Abstract

In this paper, we explore the famous graph theory game: Cops and Robbers. We first introduce fundamental concepts related to the game, formally defining cop-winning and robber-winning graphs. We then investigate the game's dynamics on famous graph families, proving lemmas that establish complete graphs and trees as cop-winning, and cycles as robber-winning. Next, we move on to define pitfalls, utilizing this concept to prove a general characterization theorem for cop-winning graphs. Following this, we define the cop number of a graph to analyze multiple-cop winning graphs. Finally, we establish an upper bound to the number of cops for any planar graph.

1 Introduction

Motivation: Imagine you are in a city in which there is a single cop actively pursuing a robber. Assume that each of them knows the layout of the city and the exact location of the other at all times, and that they take alternating turns moving from one location to an adjacent one. Can the robber escape safely and outrun the cop?

This situation can be modeled by a connected, simple, finite graph $G(V, E)$, where $V(G)$ represents all possible locations the robber and the cop can reach and $E(G)$ represent the streets. We formally define a cop, a robber, a cop-win graph and a robber-win graph.



A visual representation of the Cops and Robbers game on a city graph.

Definition 1.1. The Game of Cops and Robbers

Let $G(V, E)$ be a connected, simple, finite graph. The game of Cops and Robbers is played according to the following rules:

1. **Game Setting:** The cop chooses a starting vertex $c_0 \in V(G)$, followed by the robber who chooses a starting vertex $r_0 \in V(G)$.

2. **Turns:** The players move alternately, with the cop moving first. If a player is at vertex x, a move consists of whether they may move to any y such that $xy \in E(G)$ or remain at x.

3. **Perfect Information:** Both players have complete knowledge of the graph's structure and the other player's current position at all times.

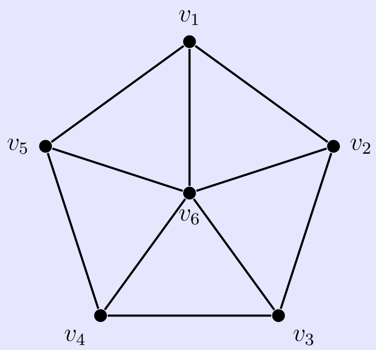
Definition 1.2

- A graph G is called a *cop-win* graph if there exists a winning strategy for the cop regardless of the robber's strategy, that is, the cop can guarantee that after some finite number of moves t, $c_t = r_t$.

- A graph G is called a *robber-win* graph if there exists a winning strategy for the robber regardless of the cop's strategy, that is, the robber can guarantee that $c_t \neq r_t$ for all turns $t \geq 0$.

Example 1.3

- All complete graphs K_n are cop-win graphs.
- All trees are cop-win graphs.
- The complete bipartite graphs $K_{m,n}$ for all $m, n \geq 2$ are robber-win graphs.
- All cycles of length greater than or equal to 4 are robber-win graphs.
- The following graph, which will be examined later, is a cop-win graph. Think about why!



Lemma 1.4

All trees and complete graphs are cop-win graphs.

Proof. We consider the two classes of graphs separately.

Complete graphs: Let K_n be a complete graph on n vertices. Since every pair of distinct vertices is adjacent, the cop may start at any vertex. After the robber chooses a starting vertex, the cop can move directly to that vertex in one move. Hence the robber is captured immediately, and so every complete graph is cop-win.

Trees: Let T be a finite tree. Since T is connected and acyclic, there is a unique path between any two vertices. For vertices $u, v \in V(T)$, denote by $d(u, v)$ the distance (number of edges on the unique path) between the two vertices u and v.

Let c_k and r_k be the positions of the cop and robber, respectively, at the beginning of round k (with $k = 1$ being the initial positions). The cop's strategy is as follows: at round k, if $c_k \neq r_k$, the cop moves to the unique neighbor c_{k+1} of c_k that lies on the c_k-r_k path. This move satisfies $d(c_{k+1}, r_k) = d(c_k, r_k) - 1$. The robber then moves from r_k to an adjacent vertex (or stays put), producing r_{k+1}.

We analyze the distance after each round. After the cop's move, the distance to the robber's current position is $d(c_{k+1}, r_k) = d(c_k, r_k) - 1$. The robber's subsequent move can change the distance by at most 1, so

$$d(c_{k+1}, r_{k+1}) \leq d(c_{k+1}, r_k) + 1 = d(c_k, r_k).$$

Thus, defining $d_k = d(c_k, r_k)$, we have $d_{k+1} \leq d_k$ for all k. The distance is a non-increasing sequence of non-negative integers.

Now suppose the robber never gets captured. Then $d_k > 0$ for all k. Since d_k is non-increasing and integer, it must eventually stabilize at some constant $m > 0$; that is, there exists K such that $d_k = m$ for all $k \geq K$. Stabilization implies that for every round $k \geq K$, after the cop's move we have $d(c_{k+1}, r_k) = m - 1$, and then the robber must move to a vertex r_{k+1} satisfying $d(c_{k+1}, r_{k+1}) = m$ (otherwise the distance would drop below m). This forces the robber to move strictly away from the cop along the unique path each round. Consequently, the robber's positions $r_K, r_{K+1}, r_{K+2}, \ldots$ form a simple path extending indefinitely without repetition. But T is finite, so this is impossible. Therefore, our assumption was false, and d_k must eventually reach 0, meaning the cop captures the robber.

Thus, one cop has a winning strategy on any finite tree.

Therefore, all trees and complete graphs are cop-win graphs. ■

Lemma 1.5

All cycles of length at least 4 are robber-win graphs.

Proof. First note that the cycle C_3 is simply the complete graph K_3, which is cop-win by the previous lemma.

Now let C_n be a cycle with $n \geq 4$. Suppose there is only one cop.

Since each vertex in C_n has exactly two neighbors, the robber can always maintain a positive distance from the cop by moving around the cycle in the opposite direction whenever necessary. In particular, because the cycle contains at least four vertices, the robber has enough space to avoid being cornered.

More precisely, after observing the cop's position, the robber chooses an initial vertex not occupied by the cop and at distance at least 2 from the cop. At each subsequent move, if the cop moves toward the robber along one direction of the cycle, the robber moves in the opposite direction. Since the graph is a cycle, this allows the robber to continuously preserve separation from the cop.

Because every vertex in C_n has exactly two neighbors and the graph forms a closed loop, the cop cannot force the robber into a trapped position. Thus, the robber can avoid getting captured indefinitely. Therefore, C_n is robber-win for all $n \geq 4$.

Hence, all cycles of length at least 4 are robber-win graphs. ■

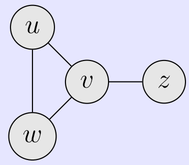
2 Cop-win graphs

Definition 2.1. Pitfall

Let $G = (V, E)$ be a simple finite graph. A vertex $u \in V$ is called a **pitfall** (or a corner) if there exists a distinct vertex $v \in N(u)$ such that its closed neighborhood is a subset of the closed neighborhood of v, i.e., $N[u] \subseteq N[v]$. In this case, we say that v **dominates** u.

Example 2.2. Visualizing a Pitfall

Consider the graph G shown to the right. Vertex u is a pitfall because its only neighbors are v and w. Notice that $N[u] = \{u, v, w\}$ and $N[v] = \{v, u, w, z\}$. Since $N[u] \subseteq N[v]$, u is a pitfall dominated by v. If the robber is at u, the cop at v can capture the robber in the next move regardless of where the robber flies.



Theorem 2.3

A finite graph is cop-win *iff* it can be reduced to a single vertex through the successive removal of pitfalls.

Proof. We prove this theorem by demonstrating both directions, following the foundational logic established by Nowakowski and Winkler [1], and independently by Quilliot [2].

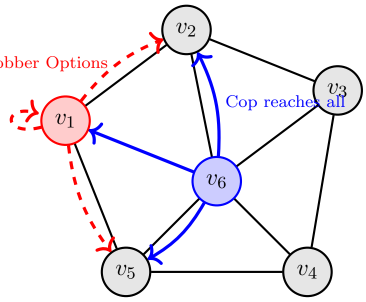
Forward Direction (⇒): If a graph is cop-win, it is dismantlable.
Assume G is a finite cop-win graph. We must prove that it contains at least one pitfall, which will allow us to start the dismantling process. Because G is finite and the cop has a winning strategy, any optimally played game must end in a finite number of moves. Let us examine the state of the board exactly *one turn before* the cop wins.

It is the robber's turn. Let the robber be at vertex u and the cop be at vertex v. Because this is the penultimate turn, the cop's strategy guarantees a capture on the very next turn, *no matter what the robber chooses to do.* Consider the robber's options. The robber can either stay at u, or move to any adjacent vertex. Therefore, the set of all possible locations where the robber could end their turn is exactly the closed neighborhood of u, denoted as $N[u]$.

For the cop (starting at v) to guarantee capture on their following turn, they must be able to reach the robber in a single move. This means v must be adjacent to every single vertex the robber could possibly occupy. Mathematically, this requires that every vertex in $N[u]$ is also contained in the closed neighborhood of v, meaning $N[u] \subseteq N[v]$. By definition, if $N[u] \subseteq N[v]$, then u is a pitfall dominated by v. Thus, any finite cop-win graph must possess at least one pitfall.

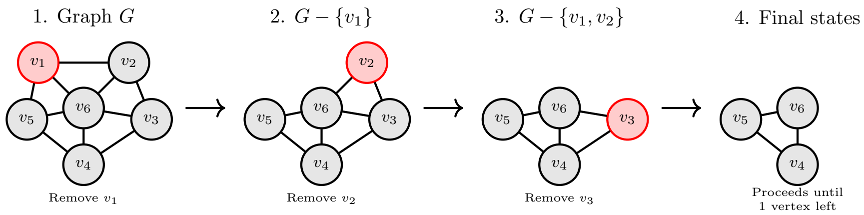
Finally, consider the subgraph $G' = G - \{u\}$. We must ensure that removing u preserves the cop-win property. While removing u strictly reduces the robber's escape paths, one might worry it also breaks the cop's winning strategy if the cop needed to traverse u. However, because $N[u] \subseteq N[v]$, any movement the cop originally planned through u can simply be rerouted through v without losing any adjacency or reach. Therefore, the cop's winning strategy on G is perfectly preserved on G', making G' cop-win as well. By repeatedly applying this logic, we can dismantle the graph vertex by vertex until only a single point remains.

Forward Direction: Penultimate Turn on W_6



Robber at v_1 has $N[v_1] = \{v_1, v_2, v_5, v_6\}$. Cop at v_6 has $N[v_6] = V(G)$.
Since $N[v_1] \subseteq N[v_6]$, v_1 is a pitfall.

Iterative Dismantling of W_6



Backward Direction ($\Leftarrow$): If a graph is dismantlable, it is cop-win.
We proceed by induction on the number of vertices, $n = |V(G)|$.

Base Case: Let $n = 1$. The graph G consists of a single vertex. The cop places themselves on this vertex and wins immediately. Thus, G is trivially cop-win.

Inductive Hypothesis: Assume that any graph with n vertices that can be reduced to a single vertex through the successive removal of pitfalls is cop-win.

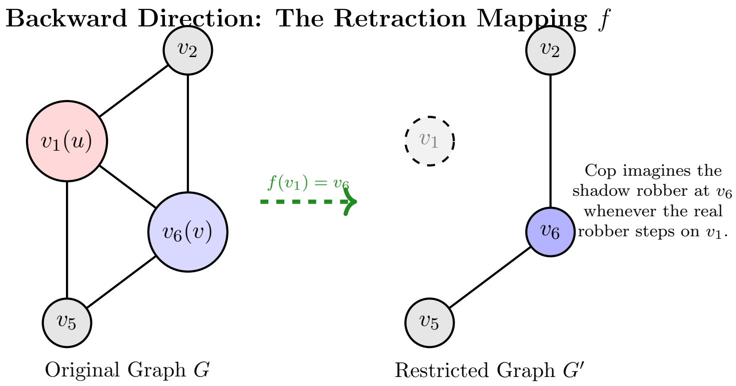
Inductive Step: Let G be a dismantlable graph with $n+1$ vertices. By definition, G must contain at least one pitfall. Let this pitfall be u, and let v be the vertex that dominates it (so $N[u] \subseteq N[v]$).

Let $G' = G - \{u\}$. Since G is dismantlable, G' is also dismantlable and has n vertices. By our inductive hypothesis, G' is a cop-win graph. We now construct a winning strategy for the cop on the original graph G using a formal retraction (or "shadow") strategy. Define a retraction mapping $f : V(G) \to V(G')$ such that $f(x) = x$ for all $x \neq u$, and $f(u) = v$. Because $N[u] \subseteq N[v]$, this mapping ensures that adjacent vertices in G are mapped to adjacent vertices (or the same vertex) in G'. When the real robber moves to a vertex x in G, the cop imagines a shadow robber moving to $f(x)$ in G'. Any legal move by the real robber in G translates to a legal move (or a pass) for the shadow robber in G'.

Since G' is cop-win, the cop executes their optimal strategy on G as if they are chasing the shadow robber on G'. In a finite number of steps, the cop will capture the shadow robber at some vertex $c \in V(G')$. This means $c = f(x)$, where x is the current position of the real robber. There are two possibilities at this moment:

- If $c \neq v$, then $f(x) \neq v$, which implies $x = c$. The cop has captured the real robber.
- If $c = v$, the shadow robber is caught at v, meaning the real robber is either at v (caught) or at u. If the real robber is at u, it is their turn to move. However, because $N[u] \subseteq N[v]$, any vertex y the robber chooses to move to (or if they stay at u) is adjacent to v. On the very next turn, the cop simply moves from v to y and captures the robber.

Thus, G is cop-win.



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3 Multiple Cop graphs

Having established the structural characteristics of cop-win and robber-win graphs in the single-cop scenario, we now generalize the analysis to include multiple cops in a graph. Assume, instead of one cop, k cops, denoted by $C_1, \ldots, C_k$, are chasing the robber. The cops are placed first on G, then the robber. In the cops' turn, only one of them moves.

Definition 3.1. Cop Number

We define the cop number of a graph G, $\mathcal{C}(G)$, as the minimum number of cops that is enough to ensure that the cops can catch the robber.

Remark 3.2

Since our analysis involves only simple finite graphs, the cop number is clearly finite since $\mathcal{C}(G) \leq n$ for any graph G where $|G| = n$.

Definition 3.3. Dominating Set

A dominating set of a graph G is a set $S \subseteq V_G$ such that every vertex in G is either in S or is adjacent to at least one vertex in S.

Definition 3.4. Domination Number

The domination number of a graph G, $\beta(G)$, is the cardinality of the minimum dominating set of G.

Remark 3.5

Given an arbitrary graph G, clearly, $\mathcal{C}(G) \leq \beta(G)$.

Theorem 3.6

Any graph G with no 3- or 4-cycles and $\delta(G) \geq n$ has $\mathcal{C}(G) \geq n$.

Proof. Let C have $n-1$ cops distributed on the vertices $\{v_1, v_2, \ldots, v_{n-1}\}$. Pick a vertex w such that $w \notin \{v_1, \ldots, v_{n-1}\}$ (Such w exists as $\delta(G) \geq n, |G| \geq n+1$). Suppose w has l neighbors and $N(w) = \{v_1, \ldots, v_k, w_1, \ldots, w_{l-k}\}$ where $w_i \notin \{v_1, \ldots, v_{n-1}\}$. Notice that $l \geq n$ and $k \leq n-1$. Thus, $l - k \geq 1$. For the sake of contradiction, assume $\{v_1, \ldots, v_{n-1}\}$ is a dominating set in G. This means that the $N(w_i)$'s must contain one v_j. If $j \leq k$, then there exists a 3-cycle (w, w_i, v_j, w), a contradiction. So, we get that $j \geq k+1$. Also, since G has no 4-cycle, $N(w_i) \cap N(w_m) = \{w\}$ for $i \neq m$. So, every w_i must be adjacent to a unique v_j. Counting the total number of vertices in the dominating set, we get that $|\{v_1, \ldots, v_{n-1}\}| \geq k + (l-k) = l$. Thus, $n - 1 \geq l$, a contradiction.

Hence after placing the cops, say $\{C_1, \ldots, C_{n-1}\}$, R is able to place himself on a vertex r which is neither equal to nor adjacent to any of the C_i's. But R can keep up this situation because, after every move of the cops, at most $n-1$ of R's neighbors are covered by cops (no 3- or 4-cycles), thus allowing R to go to the free neighbor. ■

Lemma 3.7. Isometric Path Lemma

Let G be any graph, $u, v \in V(G), u \neq v, P = (u, v_1, \ldots, v_t = v)$ a shortest path between u and v. We assume that at least two cops are in play. Then a cop C on P can, after a finite number of moves, prevent the robber R from entering P. That is, R will be immediately caught if he moves to P.

Proof. Suppose after the robber moves from r to s, there exists a vertex $z_0 \in V(P)$ such that

$$d(s, z_0) = d(c, z_0) - 1.$$

Then the cop may move one step along the path P toward z_0, thereby decreasing his distance to z_0 by 1. Hence, the condition

$$d(c, z) \leq d(r, z) \qquad \forall z \in V(P)$$

is preserved. Observe that if the robber moves onto some vertex $z \in V(P)$, then

$$d(r, z) = 0.$$

Hence,

$$d(c, z) \leq d(r, z) = 0,$$

which implies that $d(c, z) = 0$. Therefore, the cop is already on the vertex z, and the robber is immediately caught. Therefore, for the robber to be truly threatening, there must exist vertices $x, y \in V(P)$ lying on opposite sides of the cop c, say $x < c < y$, such that

$$d(s, x) = d(c, x) - 1$$

and

$$d(s, y) \leq d(c, y),$$

or vice versa. In other words, the robber would force the cop to defend one side of the path while simultaneously threatening the other side.

However, this is impossible. Since P is a shortest path and c lies between x and y on P, we have

$$d(x, y) = d(c, x) + d(c, y).$$

Moreover, by the triangle inequality,

$$d(x, y) \leq d(s, x) + d(s, y).$$

Using the assumptions above, we obtain

$$d(s, x) + d(s, y) \leq (d(c, x) - 1) + d(c, y).$$

Hence,

$$d(x, y) \leq d(c, x) + d(c, y) - 1 = d(x, y) - 1,$$

which is a contradiction.

Therefore, such vertices x and y cannot exist. Consequently, the robber can never make a threatening move, and the cop can always move appropriately along P so as to preserve the condition

$$d(c, z) \leq d(r, z) \qquad \forall z \in V(P).$$

It follows that if the robber ever enters P, then he is immediately caught. ■

Before stating the main theorem of this section, we recall the classical Jordan Curve Theorem, which we will invoke in the proof. The following statement, and a complete proof, can be found in Tverberg [3].

Theorem 3.8. Jordan Curve Theorem [3]

Let Γ be a Jordan curve in the plane, that is, the image of the unit circle under an injective continuous map into $\mathbb{R}^2$. Then $\mathbb{R}^2 \setminus \Gamma$ is disconnected and consists of exactly two connected components: one bounded (the *interior*) and one unbounded (the *exterior*).

Definition 3.9. Robber Territory

In a graph G with the robber R at the vertex r and the cops at the vertices $c_1, c_2, \ldots, c_n$, the robber territory is the set of vertices that the robber can reach without crossing the current cop-occupied vertices or cop-controlled isometric paths.

Theorem 3.10

Any planar graph G has $\mathcal{C}(G) \leq 3$.

Proof. Assume G is planar. We aim to assign to the robber in each step i a subset of V_G, say R_i, which is the current robber territory, showing that after a finite number of cop-moves, R_i can be reduced to R_{i+1} where $R_{i+1} \subset R_i$. Hence, we decrease the robber territory until, eventually, there is no vertex left for the robber to go to.
Initially, all 3 cops C_1, C_2 and C_3 occupy the vertex z_0 and the robber R occupies the vertex r. Thus, R_0 is defined to be the vertices of the connected component containing r in $G - z_0$. Inductively, at stage i (after R's move), since G is planar, one of the following two cases arises.

- **Case A:** Some cop C_1 is at vertex u, R is at r and R_i is defined to be the vertices of the connected component containing r in $G - u$ (Same situation as step 0). The other two cops are free.
- **Case B:** The robber territory R_i is bounded by two isometric paths P_1 and P_2 controlled by cops C_1 and C_2 respectively, which meet at a vertex u. The third cop C_3 is free.

We will now prove that the condition holds in **Case (A)**.
Suppose u has only one neighbour v in R_i. Then, C_1 would move from u to v. If $v = r$, then game was over. If not, no matter how R moves next, $R_{i+1} = R_i - \{v\}$. Hence, we are again back at case A with $R_{i+1} \subset R_i$ as desired.
If u has more than one neighbour in R_i, pick any two neighbours a and b. Let P be the

shortest path between a and b in R_i. By Lemma 3.7, another free cop C_2 controls P after a finite number of moves. Thus, we reach case B where cop C_1 controls $P_1 = (a, u, b)$ and cop C_2 controls $P_2 = P$ and $R_{i+1} \subseteq R_i - V_P \subset R_i$. Thus, the condition still holds as desired.

We now address **Case B**.

- **Case B(i):** Suppose there is no path in $R_i \cup P_1 \cup P_2$ from u to v other than P_1 and P_2. Then R_i consists of disjoint components $A, B, C, \ldots$ attached to vertices of $P_1 \cup P_2$. Let r be contained in component A, attached to a vertex a on, say, P_1. The free cop C_3 moves to a (while C_1 and C_2 maintain control of P_1 and P_2 respectively), and we are back to **Case (A)** with $u = a$ and $R_{i+1} = A$, so that $R_{i+1} \subsetneq R_i$ as required.

- **Case B(ii):** Suppose that the robber territory R_i has vertices adjacent to both P_1 and P_2. Let $x \in V(P_1)$ and $y \in V(P_2)$ be vertices that have neighbours inside R_i, chosen so that the segment of P_1 from u to x and the segment of P_2 from u to y contain no other vertex adjacent into R_i. Let Q be a shortest path in $G[R_i \cup \{x, y\}]$ from x to y whose internal vertices lie strictly within R_i.

  By the Isometric Path Lemma (Lemma 3.7), the free cop C_3 can, in a finite number of moves, take up a position on Q and thereafter prevent the robber from crossing Q without being immediately captured.

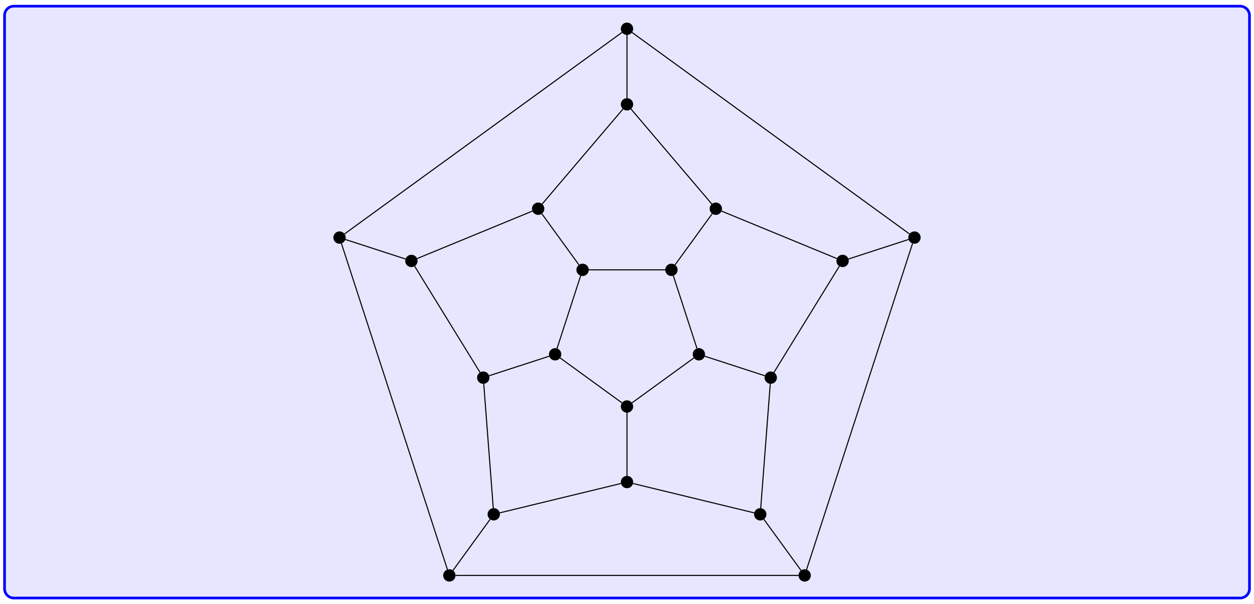
  Consider the closed walk W formed by: the segment of P_1 from u to x, the path Q from x to y, and the segment of P_2 from y back to u. Since G is planar and the three constituent paths share only their endpoints, W traces a simple closed curve in any planar embedding of G. By the Jordan Curve Theorem (Theorem 3.8, [3]), this curve divides the plane into exactly two connected regions. The robber, who cannot cross Q (guarded by C_3), the segment ux on P_1 (guarded by C_1), or the segment uy on P_2 (guarded by C_2), is confined to one of these two regions. Define R_{i+1} to be the vertices of G lying in the region that contains the robber's current position r. Since Q cuts strictly through R_i and all internal vertices of Q belong to R_i, we have $R_{i+1} \subsetneq R_i$.

  It remains to verify that the resulting configuration satisfies one of our two inductive cases and that a free cop is maintained. The new territory R_{i+1} is bounded by two isometric paths meeting at a single vertex: the segment of P_1 from u to x (still controlled by C_1) and the path Q (now controlled by C_3), meeting at the vertex x. This is exactly the structure required by **Case (B)**, with the new meeting vertex being x (not u). Cop C_2, whose path P_2 no longer forms part of the boundary of R_{i+1}, is thereby freed. Thus we return to **Case (B)** with one free cop and $R_{i+1} \subsetneq R_i$, as required.

Since in all cases we can strictly reduce the robber's territory while maintaining the required number of free cops, the induction process continues until the territory is empty and the robber is caught. Therefore, $\mathcal{C}(G) \leq 3$. ■

Example 3.11

Consider the following graph. Notice that $\delta(G) = 3$ and the graph is planar. Also, it contains no 3 or 4 cycles. So by theorems 3.10 and 3.6, $\mathcal{C}(G) = 3$, proving that our bound for cop number is tight.



4 Conclusion

In this paper, we explored the fascinating dynamics of the Cops and Robbers game on various graphs. We began by establishing that complete graphs and trees are cop-win, while cycles of length at least four are robber-win. We then presented a complete characterization of cop-win graphs, showing that a graph is cop-win if and only if it is dismantlable through the successive removal of pitfalls. Finally, we extended our analysis to the multiple-cop scenario, introducing the cop number of a graph. We proved that for graphs with no 3- or 4-cycles, the cop number is bounded below by the minimum degree, and used the Isometric Path Lemma alongside the Jordan Curve Theorem to establish that the cop number of any planar graph is at most 3. These foundational results highlight the profound connections between gameplay strategies and graph structural properties.

References

[1] Nowakowski, R., and Winkler, P., *Vertex-to-vertex pursuit in a graph*, Discrete Mathematics, 43 (1983), 235–239.

[2] Quilliot, A., *Jeux et pointes fixes sur les graphes*, Thèse de 3e cycle, Université de Paris VI, 1983.

[3] Tverberg, H., *A proof of the Jordan curve theorem*, Bulletin of the London Mathematical Society, 12 (1980), 34–38. Available at: `https://webhomes.maths.ed.ac.uk/~v1ranick/jordan/tverberg.pdf`

[4] Aigner, M., and Fromme, M., *A game of cops and robbers*, Discrete Applied Mathematics, 8 (1984), 1–12.

[5] `https://www.youtube.com/watch?v=9mJEu-j1KT0`